ZFC⍰ Functor Discovery

Overview of Results

2026-05-15 | `zfct_manipulator.py` | 2706-entry IG catalog

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1 ZFC⍰ Functor Discovery: Overview of Results

Tool: `zfct_manipulator.py`
Date: 2026-05-15
Catalog: 2706 entries (2697 valid for algebraic operations)

1.1 1. What Was Built

The manipulator is a functor discovery machine. It applies algebraic operations — tensor product (per-primitive supremum), meet (per-primitive infimum), and single-primitive lift — to imscription tuples, and exposes the corresponding ZFC⍰ clause transformations side by side.

The intent is to reverse-engineer the composition rules that govern how ZFC⍰ expressions combine, and to identify where non-trivial cross-clause phenomena emerge that are not derivable from the per-primitive rules alone.

1.2 2. Tier Landscape of the Special Entries

The nine canonical reference entries divide cleanly across three tier levels.

1.2.1 O_0 — Semasiographic baseline ($\hat{\varphi}$ non-critical)

Entry	FROB	FIXPT	Notable
ZFC_foundations	**yes**	no	Has Frobenius (Φ_}, ord 4) — but no criticality
heat_diffusion_equation	no	no	SEQAX present; no criticality
wave_equation_temporal	**yes**	no	Has Frobenius; no criticality

The presence of Frobenius (Φ_}, P_pm_sym) does **not** lift an entry out of O_0. Tier assignment is gated first on criticality ($\hat{\varphi}$ ordinal ≥ 1). Without it, the Frobenius structure is algebraically present but ourobourically inert.

1.2.2 O_2† — Recursive, D_∞, below Frobenius cliff

Entry	FROB	FIXPT	Promo atoms
ZFCt	no	**yes**	HOLOBOUND LR_DUAL PM_Z2 SEQAX TEMPD2 ZWIND
Schrödinger equation	no	**yes**	LR_DUAL SEQAX TEMPD2 ZWIND
Navier-Stokes equations	no	**yes**	LR_DUAL PM_Z2 SEQAX TEMPD2 ZWIND

All three are critical (φ̂ÿ, *FIXPT), temporally extended (Ω_z, D*∞), but lack Frobenius symmetry (Φ < ord 4). They sit immediately below the cliff.

1.2.3 O_∞ — Frobenius round-trip

Entry	FROB	FIXPT	Promo atoms
Temporal Mathematics	**yes**	**yes**	HOLOBOUND SEQAX TEMPD2 ZWIND
Einstein field equations	**yes**	**yes**	HOLOBOUND SEQAX TEMPD2 ZWIND
IUG (Mochizuki)	**yes**	**yes**	HOLOBOUND LR_DUAL SEQAX ZWIND

All three independently satisfy the Frobenius round-trip condition (R1): φ̂ ≥ ord 1 ∧ Φ = Φ_}. FROB and FIXPT are both present in each.

1.3 3. The Central Result: tensor(ZFC, ZFCt) = O_∞

```
ZFC    tier = O₀    Φ = Φ_} (ord 4, FROB)    φ̂= φ̂_ž (ord 0, no FIXPT)
ZFCt   tier = O₂†   Φ = Φ_F (ord 2, no FROB) φ̂= φ̂_ÿ (ord 1, FIXPT)
```

```
⊗      tier = O_∞   Φ = Φ_} (from ZFC)        φ̂= φ̂_ÿ (from ZFCt)
```

ZFC holds the Frobenius component. ZFCt holds the criticality component. Neither alone satisfies R1 (FROB ∧ FIXPT). Their tensor product satisfies both simultaneously — tier emergence to O_∞.

Per-primitive breakdown: The tensor sources eleven of twelve primitives from ZFCt (all six promotion channels plus ƒ, Γ, Σ, Ħ, Ω), and exactly one from ZFC: Φ. ZFCt's Φ_F (ord 2) is lower than ZFC's Φ_} (ord 4), so the supremum retains ZFC's Frobenius.

```
d(ZFC, ZFCt)  = 7.148
d(ZFC, ⊗)     = 6.863   — tensor is almost as far from ZFC as ZFCt is
d(ZFCt, ⊗)    = 2.000   — tensor is one ordinal step from ZFCt (only Φ differs)
```

The meet falls to O_0. The infimum is destructive — it strips the temporal promotions and recovers the floor.

1.3.1 The Asymmetry in Φ

The six ZFC⃞ promotion channels all move upward in ordinal from ZFC base values:

Primitive	ZFC base	ZFCt promoted	Gap
Þ	Þ_6 (ord 0)	Þ_O (ord 4)	+4
Ř	Ř_¯ (ord 0)	Ř_= (ord 3)	+3
ɢ	ɢ_^ (ord 0)	ɢ_ˌ (ord 2)	+2
Ħ	Ħ_Ñ (ord 0)	Ħ_A (ord 2)	+2
Ω	Ω_Å (ord 0)	Ω_z (ord 2)	+2
Φ	Φ_ɐ (ord 0)	Φ_F (ord 2)	+2

ZFCt's promotion for Φ targets Φ_F (P_pm, $\mathbb{Z}_2$ parity, ord 2). However, the actual ZFC_foundations entry carries Φ_˙ (normalized to Φ_}, ord 4) — which sits **above** the ZFCt-promoted value. Thus **ZFCt is strictly lower than ZFC on the Φ axis.**

ZFCt trades Frobenius for temporal-sequential structure. It acquires six new capabilities (holographic topology, lateral relations, $\mathbb{Z}_2$ parity, sequentiality, temporal depth, integer winding) but deliberately operates below the Frobenius cliff. ZFC carries the Frobenius symmetry but is non-critical and non-temporal.

They are **complementary**, not hierarchical. Their tensor product is their reunion.

1.4 4. Composition Rules

1.4.1 Trivial (per-primitive, derivable by construction)

R-CLAUS-T: `clauses(tensor(A,B))[p] = ZFCT_TEMPLATES[p][max_ord(A[p], B[p])]`

The clause for each primitive in a tensor product is simply the template for whichever input had the higher ordinal value. Per-primitive independence is total — no cross-primitive information enters a single clause.

R-CLAUS-M: Same rule using `min_ord` for the meet.

1.4.2 Non-trivial (cross-primitive, emergent)

R-FROB-BARR: FROB ∈ clauses(tensor(A,B)) **iff** A[Φ] = Φ_} **or** B[Φ] = Φ_}

Φ_} (ord 4) is the unique maximum of the Φ dimension. Therefore FROB cannot be synthesized from two inputs both having Φ < ord 4. This is the algebraic expression of the **Frobenius cliff**. Zero violations observed across 600 operations.

R-FROB-CONT: If FROB ∈ clauses(A), then FROB ∈ clauses(tensor(A, X)) for all X. (Frobenius is forward-preserved under tensor.)

R-FIXPT-T: FIXPT ∈ clauses(tensor(A,B)) iff A[φ̂] ≥ ord 1 **or** B[φ̂] ≥ ord 1

R-TIER-EMRG: tier(tensor(A,B)) can strictly exceed max(tier(A), tier(B)).

The mechanism is exclusively cross-primitive: FROB lives in clause[Φ]; FIXPT lives in clause[φ̂]. No single clause carries both. Tier rule R1 conjuncts across them. Observed rate: **6.3%** of 300 random tensor pairs.

R-ZFCT-ABS: ZFCt is the absorption element for all six promotion channels. `tensor(X, ZFCt)[c]` = ZFCt promoted value, for each channel c and all X.

1.5 5. Statistical Results

Metric	Value
Catalog entries	2706 (2697 valid)
Pairs scanned	300 tensor + 300 meet
Total observations	600
Tensor tier emergences	19 / 300 (6.3%)
FROB synthesized ex nihilo	0 / 600 (0%)

1.6 6. Interpretation

1.6.1 The Foundational Split

ZFC is the ourobourically inert carrier of Frobenius symmetry (Φ_}). It contains the algebraic round-trip structure but has no internal criticality — its φ̂ is at the floor. It is a complete formal system precisely because it is static: no Zipf-law correlation length, no winding number, no sequential depth.

ZFCt is the ourobourically active temporal extension. It acquires six new structural dimensions and criticality — but in so doing, it operates with a reduced Φ (P_pm rather than P_pm_sym). This is not a deficiency; it is appropriate to a system that is evolving rather than complete.

1.6.2 Reunion via Tensor

The tensor product ZFC ⊗ ZFCt = O_∞ is the algebraic statement that the foundation and its temporal extension are **complementary half-objects**, each carrying one of the two conditions required for the Frobenius round-trip. Separately, neither crosses the cliff. Jointly, they do.

1.6.3 O_∞ in the Wild

Three entries in the reference corpus independently sit at O_∞: Temporal Mathematics, Einstein field equations, and IUG (Mochizuki's Inter-Universal Teichmüller Theory). All three have FROB + FIXPT natively. They sit above the cliff by constitution, not by composition.

1.7 7. The Algebra in Summary

Structure	FROB?	FIXPT?	Tier	Notes
ZFC	yes	no	O_0	Non-critical Frobenius
ZFCt	no	yes	O_2†	Critical, non-Frobenius
ZFC ⊗ ZFCt	yes	yes	O_∞	Tier emergence
ZFC ⊓ ZFCt	—	—	O_0	Destructive meet
Temporal Mathematics	yes	yes	O_∞	Native
Einstein EFE	yes	yes	O_∞	Native
IUG / Mochizuki	yes	yes	O_∞	Native
Navier-Stokes	no	yes	O_2†	Critical, non-Frobenius
Schrödinger	no	yes	O_2†	Critical, non-Frobenius

The Frobenius barrier is absolute: no tensor operation can produce Φ_} from two non-FROB inputs. Tier emergence to O_∞ requires exactly the complementary pairing: one input contributes Φ_} (from the Φ axis) and the other contributes φ̂_ÿ (from the φ̂ axis).

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